ENGINEERING SCIENCE

Revised June 2019

Pupil booklet

Context and challenges:

Outcomes 2 and 3 only

Topic 1: Investigate engineered objects

Topic 2: Investigate engineered challenges and relate these too key engineered concepts.

Topic 3: Describe the impacts of engineering.



Name: _____

Class: _____

Teacher: _____

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Outcomes and Assessment standards

The aim of this unit is to develop a basic understanding of engineering, its role and impact on our society and environment. There are main 3 outcomes in this unit and it has been divided into two booklets. By the end of this first booklet, you will have completed outcomes 2 and 3. You will investigate engineering systems, problems and solutions, involving some existing and emerging technologies and consider implications relating to the environment, sustainable developments, and to economic and social issues.

The unit outcomes are:

1. Investigate engineered objects.
2. Investigate engineered challenges and relate these too key engineered concepts.
3. Describe some aspects of the impact of engineering.

Outcome 1

The candidate will:

1 Investigate engineered objects by:

1.1 Describing, using the systems approach, how some engineered objects work.

1.2 Identifying sub-systems and describing the function of each and how they interact.

1.3 Producing system diagrams to show sub-systems.

1.4 Carrying out energy audits.

Outcome 2

The candidate will:

2 Investigate engineering challenges and relate them to key engineering concepts by:

2.1 Identifying and describing how several different branches of engineering contribute to solving an engineering challenge.

2.2 Describing examples of the varied roles of engineers in designing, implementing, testing and controlling complex systems.

2.3 Modelling some aspect (related to one branch of engineering) of a solution to an engineering challenge.

2.4 Explaining how emerging technologies may provide improved solutions to engineering challenges.

Outcome 3

The candidate will:

3 Describe some aspects of the impact of engineering by:

3.1 Describing examples of social and economic impacts of engineering.

3.2 Describing some examples of environmental impacts of engineering.

3.3 Describing some ways in which engineering solutions contribute to tackling climate change.

Success criteria

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

☐ I understand the role of various engineers within design projects.

☐ I can identify the differences between skills and knowledge used by a variety of engineering disciplines.

☐ I am able to debate the possible future impact of new and emerging technologies one economic prosperity and the environment.

☐ I am able to discuss possible lifetime costs of the environment in Scotland and Beyond.

Why is Engineering important?

Engineering is all around us and it is important that we understand the wider role of engineering in society. In this unit you will learn about the role of different engineering disciplines. Likewise, you will also learn about one of the greatest engineering challenges that faces us today - energy. You will likely already be aware of the role of renewable energy as part of a developing society. In this unit you will further that understanding through studying energy changes and inefficiencies which will enable you to audit and explore existing systems.

What is 'Technology'?

1. Copy the following statement into your jotter and insert the missing words from the following:

Mechanisms ***Computer Control*** ***Energy*** ***Pneumatic Systems*** ***Electronics*** ***Structure***

It does not matter if it is the design of the bus to the latest mobile phone, it all depends on technology and the role of an engineer. A structural engineer will have an input in the construction of a house/office by a knowledge of the loads the _____ can withstand. Without the knowledge of _____, nothing would move. A heating system in is controlled using _____ and the factories where food is manufactured will use _____ _____ and _____ _____. All of these areas of technology need to be powered _____ and are going to be studied during your Engineering Science Course.

Definition of technology



The World we live in is a much more comfortable place. Without technology and engineering tools for hunting and fire for heating/cooking, human civilization would still be in the Stone Age.

You will quickly see that all fields of Engineering rely on Technology. Technology is the combination of knowledge and resources to solve problems.

Today, more than ever, Technology is rapidly developing. This can be seen in the mobile phone market in addition to music players, cars and games consoles.





The images opposite show how fast 3D graphic technology and processing technology has moved on. The image on the left is from first Tomb raider which was released in Japan in 1996 and right is from a format released in 2018. Take 1 minute to consider what technology you might experience / look like in the next ten years.

2. Your task is to pick an everyday piece of technology and create a timeline for it. It will require online research. In your jotter and with the use of images

glued to support your timeline, explain how it started and it has progressed throughout the years. For example, consider the creation and development of: audio (Morse code/FM/DAB), visual (film/television), transport- (industrial/commercial/personal) and communication.

3. Read the following statements. Write each statement into your jotter and one put a T or F to say whether you think it is **TRUE** or **FALSE**.

- Technology is only about electronics.
- Technology is about solving problems.
- Cars use technology but windmills don't.
- People have always used some form of technology.
- The World would be a better place without technology.
- Everybody is affected by technology.
- Only scientists use technology.
- Cavemen did not use technology.



4. Copy and complete the sentence below by putting each of the 5 words listed in an appropriate place in order to reach a definition of technology that we will use for this course.

LIVES ***RESOURCES*** ***MAKING*** ***PROBLEMS*** ***KNOWLEDGE***

Technology is about …… things by using …… and …… to solve …… in, and attempt to make our …… better.

Engineering: The role of the engineer



5. Investigate the role for each of the following seven engineers and write a few paragraphs which describes the roles of different Engineers within a variety of Engineering Challenges. Below is an engineering definition example for Civil Engineering.

Electronic ***Electrical*** ***Civil*** ***Mechanical*** ***Chemical*** ***Structural*** ***Environmental***

"One of the oldest and largest branches of engineering is Civil engineering. Civil engineering involves knowledge of working on land and water. They are experts in designing and assisting in the operation and maintenance of water resources, bridges, railways, tunnels, transportation systems, environmental protection systems and urban development. Without civil engineers our society would not be the way it is today.

The appearance and operation of structures like sky scrapers, offices blocks and even airports will be created by Architects but it will be a team of Civil engineers that will deliver the project along with many other engineers. However, some projects will not need architects and only Civil engineers to create them such as renewable wind farms, water treatment works and road infrastructure upgrading. The role of the Civil engineer will involve many aspects such as planning and project costing management, designing, Health and Safety, construction quality control, material ordering, setting out, working with clients and the public, provide daily duties to the physical construction labour and supervising construction."

One of Civil engineering's greatest accomplishments is the Great wall of China. It stretches more than 6,000 miles long. Without the correct geological research, planning, material and following the correct procedure of construction, it would not have stood the test of time. Major UK civil engineering companies are Balfour Beatty and Sir Robert McAlpine and Morrison.



6. Although there is a huge array of paths that an engineering career could lead you, there are four main branches within Engineering: Civil, Mechanical, Electrical and Chemical. These can then be broken

down further into more complex specialism careers paths. Go to **myworldofwork.co.uk** and then select '**my career options**' and search for 'engineering'.

Using the sample list of specialist engineers provided and the 'sample job reference sheet' on the next two pages, copy and complete a table and questions for at least four specialist types of engineer. Could one of these jobs be your future career? What education qualifications to I need to study Engineering at University?

When answering a question about the '*role of an engineer*', it is important to be specific to the context the question is given in. The following table will assist you in identifying specific types of engineer's **knowledge** and **skills**. There is and Context and Challenge exam question and answer at the bottom of page 8.

Type of Engineer	**Knowledge**	**Skills**
Electronic Engineer / Electrical Engineer Similar in name **but** different how?	Understanding of how each electronic component works. Understanding of how electronic circuits work.	Design electronic sub-systems. Program IC chips. Design and analyse electronic circuits.
Civil	Knowledge of properties of materials to influence dimensions and maximum loads. Knowledge of joining techniques.	Design and build bridges, roadways, train lines, tunnel… Decide on materials to use for each of their projects.
Mechanical	Knowledge of properties of materials. Knowledge of different mechanical components and their purpose. Knowledge of shapes and forms to create a stable structure.	Design mechanical sub-systems. Analyse existing mechanisms and suggest improvements. Design supporting structures for mechanisms.
Chemical	Knowledge of properties of chemicals and materials. An understanding of energy changes within selected components.	Analyse energy changes within a system. Design systems to be compatible with certain fuel types and suggest ways to harness this fuel more effectively.
Structural	Material properties such as ultimate stress and strain values for each material, brittleness, ductility. Knowledge of shapes and forms to create a stable structure.	Design structures to cope with certain forces, stresses and strains. Choose materials suitable to different structures.
Environmental	Use the principles of **engineering**, soil science, biology, and chemistry to develop solutions to **environmental** problems.	They are involved in efforts to improve recycling, waste disposal, public health, and water and air pollution control.

Below is a sample of the various types of engineers taken from the website called My World of Work.co.uk

Aerospace engineer	Agricultural engineer

Automotive engineer	Broadcast engineer
Biotechnologist	Chemical engineer
Clinical engineer	Civil engineer
Design Engineer	Drilling Engineer
Electrical engineer	Electronics engineer
Energy engineer	Geneticist
Material engineer	Marine engineer
Manufacturing systems engineer (Process engineer)	Measurement & control engineer (Geotechnical engineer)
Mechanical Engineer	Medical physicists
Structural engineer	Network engineer (power system engineer)
Physicist	Software developer
Studio sound engineer (acoustic)	Telecoms engineer
Watch or clock repairer	Geotechnical engineer
Computer engineer	Environmental engineer
Thermal engineer	Molecular engineer
Transport engineer	Technician engineer (civil)

My world of work: Sample job research sheet

What is the job title?	
What are the main duties of this job?	
What are the skills required?	

What qualifications do you need?	
What have you learned in your school subjects that might be useful in this job?	
What key points about this job do you want to share with the rest of the class?	

Exam question example:

A wind farm is being designed in a rural area of the highlands. Explain what is meant by 'specialist knowledge' and 'specialist skills' which a civil engineer could contribute to the design.

Civil Engineer **Knowledge**: A civil engineer would need to know (Via university studies) about different materials and how they work together. They would have to know about project management, Health and Safety regulations, contract law and the complex calculations required to set out a project. They need to learn and understand many aspects of Civil engineering before they can apply. Knowledge of how survey equipment operates and the complex calculations needed to be input into the machine. Only then can a road direction, concrete foundation position level or drainage pipe levels be identified.

Civil Engineer **Skill**: A civil engineer could design and construct a new road network. Questions of how is the power being transferred to the power station must be solved Their role may including the planning and implementation of bridges or stream diversions to allow access to the wind farm. They would be responsible for Health and Safety on the site.

Engineering: Context and Challenges report

Investigating an engineering challenge and how to plan a report



7. You are required to investigate and create a two-part report on an engineering project. You could do one of the examples listed below:

The Queensferry Crossing, The Falkirk wheel, HMS Queen Elizabeth aircraft carrier, The Edinburgh Trams, The new James Gillespies' High School, The Scottish Parliament, A9 Duling, M8 extension, London

underground cross rail network, Off shore wind farms, Amazon distribution warehouse (Dunfermline) or any other engineering challenge. Your teacher will show you an engineering challenge presentation report.

Part 1 for the context for your report should include the following SEVEN sections and possible evidence.

Section	Types of engineers required?	evidence
1. Context	▪ What was the context for the engineering challenge? What is it? ▪ What was the background leading up to the **need** for the project? Purpose?	Any related images / photographs? Cost? Project duration? Location?
2. Engineering context	**for example:** ▪ Civil/Mechanical engineer ▪ Electrical engineer ▪ Chemical engineer	Explain what role (skills and knowledge) would each of these engineers bring to the engineering project such as this? Provide at least 3 of the 7 Engineers roles for skills and Knowledge they would bring to the project (reference task 6 for help).
3. Engineering solution	**for example:** ▪ Civil/Mechanical engineer ▪ Electrical engineer ▪ Chemical engineer	What innovations or new technologies did any of these engineers contribute specifically to the project you have chosen.

Impacts of engineering

Part 2 of your report which focuses on 'The impacts of engineering'. Within any engineering challenge or solution, we also have to consider the impact on the population and the surrounding areas, which can be positive as well as negative.

You must answer the following questions:

a) Describe at least **one positive and one negative social impact** resulting from an engineering project or challenge.
b) Describe at least **one positive and one negative economic impact** resulting from an engineering project or challenge.
c) Describe at least **one positive and one negative environmental impact** resulting from an engineering project or challenge.

Below is an example of some questions you might need to consider when investigate an engineering challenge and context report.

Social impacts

– Is there increased employment or training opportunities created through this project?
– Will there be improved infrastructure because of it?

– Will there be traffic disruption because of its creation?
– Will there be disturbances because of noise?

Economic Impacts

– Will this solution bring money into the local area through tourism or other means?
– Will it attract other companies to invest in the area?
– Will it employ more people, meaning more money spent in the local area?

Environmental Impacts

– Could this provide habitats for wildlife?
– Will there be a risk of damaging animal habitats or ecosystems?
– Will there be a loss of green belt? Will nature be destroyed/ damaged because of it?
– Will there be a risk of danger to animals because of this engineering solution?
– Will there be more demand on water or power services?

Use the table below to help you consider some other questions for your report.

4. Social impact	▪ Changes to living conditions ▪ Infrastructure changes, housing, traffic ▪ Population movement
5. Economic impact	▪ Overall cost of the project ▪ Job creation ▪ National and local impact
6. Environmental impact	▪ Geology, visual impact, air quality, wildlife pollution ▪ Material usage. ▪ Environmental agencies, RSPB, SEPA, Historic Scotland etc.
7. Environmental alternatives	▪ Investigate alternatives to energy sources for the project. ▪ Alternative materials?

An SQA sample report is provided at the end of this booklet.

Emerging technologies

Another thing that engineers have to consider when coming up with a solution is emerging and new technologies. Is there a new or emerging material that would be more suitable for the task? Is there a new process for completing the task that will be better suited for the job? Has something been created that makes it more economical or efficient? If it is powered, could it be run on a different type of fuel that is more environmentally friendly?

Emerging technologies are technologies that are perceived as capable of changing the status quo. Emerging technologies as of 2016 include a variety of technologies such as educational technology, information technology, nanotechnology, biotechnology, cognitive science, psychotechnology, robotics, and artificial intelligence. Graphene is a material that has the potential to revolutionise a huge number of areas from electronics to energy production and structures.

By searching online for 'Emerging Technologies', you will find lots of information out there. If you are struggling, the use of website 'Scientific America' and the 'World Economic Forum'. It creates a top ten list every year of emerging technologies. Another thing engineer's need to consider is how a new emerging technology or material that would be more suitable if used in a different way or product. Would the operation of a product by using an alternative fuel type for example make it more economical, environmentally friendly or efficient?



7. Engineers often make use of emerging technologies to develop better solutions. Some emerging technologies have been provided but you can investigate your own. Investigate two and explain what some of these technologies are and how they are used.

- wireless power transmission
- touch sensitive screens
- OLED's
- Smart glass
- near-field communication

Here is a brief example of how to these technologies are used:

'Wireless power transmission would mean that I could charge my phone without plugging it in.'

'Graphene could be used to make the structural parts of things we use as well as the electronics that control them.'

'OLED screens could be shaped around lots of different objects rather than just being flat.'

A sample SQA answer is provided to illustrate the level and quality of response expected. The report must have 8 clear sections as specified earlier. Section 8 of 'emerging technologies' can be answered in a separate report.

New Forth Crossing

Main branches of Engineering involved in this project are Civil, electrical and electronic, structural and mechanical. During the design stage the challenge for structural and mechanical engineers was to design a bridge which had a span of more than 1.5 km but could be built for considerably less than the existing suspension bridge would cost to build at today's prices. It was decided that the bridge will be 'cable stayed' which gives strength and lightness and does not need the huge foundations needed for suspension cables.

Context of engineering

Engineers are also involved in implementing the project there are three main stages (1) putting in the road infrastructure into place to allow vehicles to get to and from the bridge from the existing roads and placing and preparing the foundations for the bridge towers this will be overseen by Civil Engineers. (2) building each of the bridge sections and then assembling them on site this will be overseen by mainly mechanical and production engineers (3) putting into place all of the control technology including computers, cameras and signs to allow the bridge traffic to be monitored and controlled this will be overseen by mainly electrical and electronic engineers.

Note: Refers to the roles of three different types of engineers

Before any engineering project can be used, it has to be thoroughly tested. Engineers will develop and implement test plans on all aspects of the bridge. Sensors embedded in the bridge will continue to supply data which is analysed continuously by engineers. Finally, engineers are involved in controlling complex electronic monitoring systems for the bridge.

Note: Refers to design, implementation, testing and control tasks

An emerging technology in this area is self-healing materials which will be able to repair damage to themselves. Once the parts have been tested they are brought to the site by ship. An emerging technology in this area is the move away from diesel engines towards hydrogen-powered ones and other alternative fuel.

Note: Refers to two different relevant emerging technologies

A sample SQA answer is provided to illustrate the level and quality of response expected.

The effects of building a bio-fuel plant in Fife

The contractors employed local companies to supply the men and equipment needed to carry out the work on the site, this provided employment and income for these companies for nearly a year. The local council were also involved in adding a roundabout and widening the road to join the access roads to the road network. Once the site was prepared another local contractor did all the steelwork required for the building structures and framework on which the specialist equipment was then added.

Social impact

During construction there was some disruption to the people staying close to the site due to noise, dust. And the many lorries, diggers etc which had to be brought to and from the site. There was also disruption on the road next to the site due to access and then road works. The money this project brought into the local economy also helped the whole Scottish economy due to taxes paid and the fact that projects like this encourage renewable energy companies to invest in Scotland. Due to projects like this Fife has become one of the centres of growth for the renewable energy sector in Europe.

Negative environmental impact

Economic impact

This project uses waste material to generate electricity, so that finite resources such as coal, gas and nuclear material is not used up. The project burns bio-gas and grain cake so there are still emissions of sulphur and nitrogen oxide and carbon monoxide and dioxide. Not only will the distillery no longer need to take power from the national grid it will actually feed extra power it generates back into the national grid. These emissions however are much less than from a power station producing the same amount of electricity. The project reduces the amount of grain wash pumped into the river forth. The firth of forth is the second most important site in the country for overwintering birds, mainly waterfowl which feed on worms on the mudflats. The grain wash will also have a cleaner added to it which will improve the water quality in the river.

Tackling climate change — weak answer here — suggest further learning and re-assessment of Assessment Standard 3.3.

environmental impact

Lesson extension overview

An activity using myworldofwork.co.uk to link the study of Engineering Science to possible careers.

Objectives

As part of a wider lesson, this lesson insert intends to:

- Encourage research into careers related to the study of Engineering Science
- Highlight careers in Engineering Science that focus on exploring engineering problems and finding solutions

My World of Work activity (20mins)

- Arrange pupils individually or groups of two or three
- Go to myworldofwork.co.uk
- Go to my career options
- Ask the groups to choose a job card from the pack
- Ask them to complete the job research sheet. Investigate several career paths.

Connect the learning (in the wider lesson)

- Discuss what each job entails and how it links to the activities they are doing in class
- Discuss which skills they are developing in class relate most closely to particular roles they have looked at

Review and reflect (5mins)

- Ask pupils to reflect on the jobs covered and if they may be interested in following one of these careers
- Homework: Ask the class to use this lesson as part of their overall research on careers related to their school subject

Resources

- Computers or tablets with internet access
- Job research sheet

Curriculum links

Suitable for all year groups studying: Engineering Science – National 4

Career Education Standard (3-18)

- Supports entitlements set out in the Career Education Standard for young people to:
- Experience a curriculum through which they learn about the world of work and job possibilities and which makes clear the strengths and skills needed to take advantage of these opportunities
- Know where to find information and access support making effective use of online sources such as My World of Work
- Develop Career Management Skills as an integral part of their curriculum